ALM7 & 5th iLS Conference
Perth, Western Australia | October 18-21, 2026

Sunday 18 October 2026

9:00 AM - 2:00 PM	**ANPC EMCR pre-conference workshop**	Workshop Australian National Phenome Centre
3:15 PM - 4:00 PM	**Conference Registration**	Registration The Mezzanine
4:00 PM - 4:30 PM	**Opening & Welcome to Country**	Opening Riverside Ballroom
4:30 PM - 5:30 PM	**Opening Plenary - Prof. Kerry-Anne Rye**	Plenary Riverside Ballroom
4:30 PM	**Lipidomics and Lipid Function-Measuring Lipids is no Longer Enough** Prof. Kerry-Anne Rye	
5:30 PM - 6:00 PM	**ECR Rapid-Fire Talks**	Session Riverside Ballroom
5:30 PM	**Quantification and Localisation of New Brain Lipid Synthesis Using Deuterium Oxide and High Resolution Mass Spectrometry** Catherine Zhang	
5:34 PM	**Circulating fatty acid oxidation intermediates predict survival in ageing men: a prospective metabolomics study** Manthan Sharma	
5:38 PM	**Investigating Microglial Lipid Metabolism in Response to Amyloid-Beta: Implications for Alzheimer's Disease** Ayesha Templeman	
5:42 PM	**The lipid landscape behind harmful extracellular vesicle release in vascular aging** Elizaveta (Eliza) James	
5:46 PM	**Lipid remodelling defines divergent cancer immunotherapy outcomes** Nicola Principe	
5:50 PM	**Integrative Modelling of Lipidomics and Lipoproteomics: A Comparative Approach for BMI Prediction** Novia Minaee	
5:54 PM	**A Systematic Evaluation of Extraction and 4D-MS Parameters for Comprehensive Human Blood Lipidomics** Nurul Saadah Ahmad	
6:00 PM - 8:00 PM	**Welcome Reception**	Social Silver Room

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Monday 19 October 2026

8:00 AM - 8:45 AM	**Registration**	Registration The Mezzanine
8:45 AM - 9:30 AM	**Keynote Prof. Hiroshi Tsugawa**	Keynote Riverside Ballroom
8:45 AM	**Toward agent-assisted lipidomics: extending the MS-DIAL ecosystem from laboratory-scale lipidomics to reproducible public data reanalysis** Prof. Hiroshi Tsugawa	
9:30 AM - 10:30 AM	**AI, Computational Tools & Data Standards for Lipidomics 1** *Computational tools and software in lipidomics*	Session Riverside Ballroom
9:30 AM	**LipiRich: An Open-Source, Browser-Based Application for Normalisation, Statistical Analysis, and Pathway Enrichment using Untargeted Lipidomics Data** Sarah Hancock	
9:45 AM	**An atlas of imputed plasma lipidomic biomarkers for health and disease in 197,840 individuals from the UK Biobank** Changyu Yi	
10:00 AM	**A pilot study of AI assisted LC-MS lipid structure identification for under represented lipid categories** Zhixu Ni	
10:15 AM	**A high-throughput LC-MS and machine learning framework for population-scale plasma lipidomics** Amy W. Liang	
10:30 AM - 11:15 AM	**Morning Tea**	Break Silver Room
11:15 AM - 12:15 PM	**Lipids in Health and Disease 1** *Membrane, in-vitro biology and biosensors*	Session Riverside Ballroom
11:15 AM	**Melanoma cells remodel their lipidome yet maintain membrane biophysical properties during drug adaptation** Aurelie Benfield	
11:27 AM	**ER translocon gating governs metabolic allocation through selective lipid routing** Carl Mousley	
11:39 AM	**Co-fractionation multiomics for mapping the composition of native membrane lipid-protein complex interactomes** Liuyu Peng	
11:51 AM	**Effects of Reactive Oxygen Species on Liposomal Biomimetics of the Mitochondrial Membranes: A Biophysical Approach** Susana Mesen-Porras	
12:03 PM	**Characterization of a lipid droplet-associated microalgal protein as a novel phosphatidic acid biosensor** Bilge Ercan	
12:15 PM - 1:45 PM	**Lunch & Poster Session**	Lunch Silver Room
POSTER 1	**High-throughput regioisomer resolved analysis of phosphoinositides** Kevin Huynh	
POSTER 3	**A Comprehensive Targeted Lipidomics Workflow for Characterisation of Multiple Biological Matrices** Dana Hicks	
POSTER 5	**Quantification and Localisation of New Brain Lipid Synthesis Using Deuterium Oxide and High Resolution Mass Spectrometry** Catherine Zhang	
POSTER 7	**Bridging Lipid Annotation and Chemometric Analysis Through a Cross-Platform Nontargeted Lipidomics Workflow** Chris Fouracre	
POSTER 9	**Circulating fatty acid oxidation intermediates predict survival in ageing men: a prospective metabolomics study** Manthan Sharma	
POSTER 11	**Clinical Translation of Targeted Lipidomics: Validation of BREASTEST plus v2 for Early Breast Cancer Detection** Mark David	
POSTER 13	**Integrative Modelling of Lipidomics and Lipoproteomics: A Comparative Approach for BMI Prediction** Novia Minaee	
POSTER 15	**A unified workflow for small molecule analysis for exposomics and molecular epidemiology** Bruker Sponsor	
POSTER 17	**Targeted lipidomics identifies markers for skin collection in a minimally invasive micro-biopsy punch** Kailyn O'Connor	
POSTER 19	**Lipidomic Characterisation of Human Monocytes, Macrophages and Monocyte-Derived Microglia-Like Cells** Matilda van Buuren-Milne	
POSTER 21	**Glass-Like Well Plates for High-Throughput Lipidomics: Limiting Analyte Loss and Reducing Extractables** Trajan Sponsor	
POSTER 23	**Characterising Coffee Lipids from Green Bean to Brew: A Lipidomic Approach** Jermaine Marie Ann Fabella	

POSTER 25	**Serum lipidomics reveals class-specific lipid remodelling and altered acyl chain composition in Parkinson's disease** Jack Price	
POSTER 27	**Ion mobility separation of lipids using a MALDI XS source with a Waters Cyclic IMS P20** Marten F. Snel	
POSTER 29	**Profiling Short-Chain Acylcarnitines in Human Peripheral Blood Mononuclear Cells** Thy Duong	
POSTER 31	**Root-shoot lipidome interactions in Medicago truncatula after drought stress** Alastair Ross	
POSTER 33	**Lipid remodelling defines divergent cancer immunotherapy outcomes** Nicola Principe	
POSTER 35	**Pre-Analytical Matrix Considerations in Sepsis Lipidomics: A Systematic Review Comparing Serum and Plasma Profiling in Bacterial Infections** Nurul Saadah Ahmad	
1:45 PM - 2:30 PM	**Keynote - A/Prof. Hyungwon Choi**	Keynote Riverside Ballroom
1:45 PM	**Scalable workflows for population-scale lipidomics in cardiometabolic diseases** A/Prof. Hyungwon Choi	
2:30 PM - 3:15 PM	**Clinical Lipidomics: From Pre-analytics to Cohort Studies 1** *Clinical and population lipidomics*	Session Riverside Ballroom
2:30 PM	**Comprehensive, multi-ancestry lipidomic GWAS reveals molecular pathways linking the human lipidome to cardiometabolic, neurodegenerative, and cancer risk** Corey Giles	
2:45 PM	**Lipoprotein and lipid perturbations induced by acute and chronic inflammation** Samantha Lodge	
3:00 PM	**A lipidomic risk score improves cardiovascular risk assessment in statin treatment across large cohorts** Jingqin Wu	
3:15 PM - 3:50 PM	**Afternoon Tea**	Break Silver Room
3:50 PM - 4:20 PM	**Platinum Sponsor Invited Talk: SCIEX**	Session Riverside Ballroom
3:50 PM	**Large panel of lipid identification and quantitation using ZT Scan DIA analysis** Dr. Paul Baker	
4:20 PM - 5:00 PM	**Clinical Lipidomics: From Pre-analytics to Cohort Studies 2** *Clinical and population lipidomics*	Session Riverside Ballroom
4:22 PM	**When recovery differs: Sex-based lipidomic changes after childhood burn injury** Monique J Ryan	
4:34 PM	**Distinct plasma lipidomic patterns across cardiovascular outcomes with contrasting profiles for stroke and heart failure** Jaya Ajay Singh	
4:46 PM	**Identification of novel lipidomic biomarkers in first trimester maternal serum for the screening of aneuploidy specifically trisomy 21.** Khoa Lam	
5:15 PM - 6:00 PM	**Australian Lipid Society AGM**	AGM Riverside Ballroom

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Tuesday 20 October 2026

8:00 AM - 8:45 AM	**Registration**	Registration The Mezzanine
8:45 AM - 9:30 AM	**Keynote - Prof. Gerald Watts**	Keynote Riverside Ballroom
8:45 AM	**Go with the flow: unravelling lipoprotein metabolism with stable isotopes** Prof. Gerald Watts	
9:30 AM - 10:30 AM	**Lipids in Health and Disease 2** *Lipoproteins and lipids in cardiometabolic disease*	Session Riverside Ballroom
9:30 AM	**Defining the potential of alkylglycerol supplementation for treating diverse metabolic disorders** Richard G. Lee	
9:45 AM	**Non-Canonical Sphingolipid Remodeling in Adipose Tissue Tracks Insulin Resistance and Type 2 Diabetes Remission** Elin Chorell	
10:00 AM	**Targeting Endothelial Plasma Membrane Cholesterol to Mitigate Atherogenesis** Jacob (Yanfei) Qi	
10:15 AM	**Integrative lipidomic analysis across six cohorts reveals biological pathways underlying cardiovascular disease risk** Tingting Wang	
10:30 AM - 11:15 AM	**Morning Tea**	Break Silver Room
11:15 AM - 12:15 PM	**Microbial & Plant Lipids in Health and Biotechnology 1** *Lipidomic applications in animals, milk and microbiota*	Session Riverside Ballroom
11:15 AM	**Free fatty acid consumption, metabolism and biosynthesis in honeybees** Sam Hodgkinson	
11:27 AM	**Untargeted lipidomics and proteomics to expand molecular coverage of human milk** Alexandra George	
11:39 AM	**High marbled Wagyu beef vs. low marbled Water buffalo beef - are there differences in the lipidome of muscle between both breeds?** Dirk Dannenberger	
11:51 AM	**Whole-grain rye feeding drives tissue-specific host-metabolic and microbial responses beyond conventional clinical endpoints** Ambrin Farizah Babu	
12:03 PM	**Interactive effects of lactation stage, dietary supplementation, and maternal metabolic state on human milk lipidome** Baoru Yang	
12:15 PM - 1:45 PM	**Lunch & Poster Session**	Lunch Silver Room
POSTER 4	**Application of Zeno SWATH DIA in multi-tissue metabolomic profiling** Dr. Thusitha Rupasinghe	
POSTER 12	**Investigating Microglial Lipid Metabolism in Response to Amyloid-Beta: Implications for Alzheimer's Disease** Ayesha Templeman	
POSTER 20	**The lipid landscape behind harmful extracellular vesicle release in vascular aging** Elizaveta (Eliza) James	
POSTER 2	**Development of a targeted lipid plasma method with a propanol free mobile phase** Lee Thomasson	
POSTER 22	**A Systematic Evaluation of Extraction and 4D-MS Parameters for Comprehensive Human Blood Lipidomics** Nurul Saadah Ahmad	
POSTER 14	**Lipidomic analysis reveals a distinct PUFA-rich profile in endometriosis** Sarah W. Walsh	
POSTER 8	**Development of a metabolomic screening algorithm for the early detection of genetic disorders using newborn dried bloodspots: The NewbornsInSA Study** Tomas Rozek	
POSTER 16	**Lipidome-wide analyses provide novel insights into the relationship between lipid metabolism, type 2 diabetes and prediabetes risk.** Habtamu Bedimo Beyene	
POSTER 6	**MStargetR: A reproducible, auditable, and containerised R workflow for end-to-end targeted mass spectrometry data processing** Harrison Szemray	
POSTER 24	**Targeting lipid and cholesterol metabolism in endometrial cancer** Claudia E. Tiplady	
POSTER 10	**Enabling Precision and Translational Lipidomics Through Bespoke Deuterated Lipids from ANSTO's National Deuteration Facility** NDF sponsor	

POSTER 18	**High-throughput Single-cell Multi-OMICS Using Standard-flow LC and a Multi-reflecting TOF MS** Marten F. Snel	
POSTER 26	**Total Lipid Extract (TLE) from Bovine Liver (Avanti) on Orbitrap Tribrid Apex Small Molecule MS** Thermo sponsor	
POSTER 28	**Sex-Specific Lipid Signatures Associated with Paracetamol Exposure** Shaimaa Khandaker	
POSTER 30	**Optimization and Clinical Validation of a Cell-Free Cholesterol Efflux Capacity Assay Using Immobilized Liposomes-bound Magnetic Beads in Statin-Treated Patients** Wanida Laiwattanapaisal	
POSTER 32	**Acetonitrile-Derived Acetamidinium Adduct Formation Reduces Signal Intensity in LC-MS Lipidomics** Nat Mellett	
POSTER 34	**Enhancing the biodistribution of artificial lipid droplets as a novel lipid particle delivery system in vivo** Ebony Monson	
POSTER 36	**Circular Bioprocessing: A study of spent media and biomass recycling with Umbelopsis isabellina** Jasmeet Kaur Khanijou	
1:45 PM - 2:30 PM	**Keynote - Dr. Laura Bindila**	Keynote Riverside Ballroom
1:45 PM	**Multi-modal 4D- and targeted lipidomic strategies for neuro- and immuno-metabolism phenotyping in health and disease** Dr. Laura Bindila	
2:30 PM - 3:15 PM	**Advancements in Lipidomic Technologies 1** *Novel lipidomic methods and lipidomic applications in tissue*	Session Riverside Ballroom
2:30 PM	**Multi-Region Brain Lipidomics in Alzheimer's Disease Neuropathology** Kevin Huynh	
2:45 PM	**Analysis of Total Fatty Acids by SFC/MS: Separation of Omega-3 and Omega-6, Branched and Hydroxy Fatty Acids** Michaela Chocholouskova	
3:00 PM	**In vivo stable isotope labelling and mass spectrometry reveals that the APOE4 is a loss of function allele for brain lipid turnover** Yushan Cai	
3:15 PM - 4:00 PM	**Afternoon Tea**	Break Silver Room
4:00 PM - 5:00 PM	**Advancements in Lipidomic Technologies 2** *Novel lipidomic methods and lipidomic applications in tissue*	Session Riverside Ballroom
4:00 PM	**Rapid, High-Throughput Lipid Class Separation Using Supercritical Fluid Chromatography-Mass Spectrometry (SFC-MS)** Rangika Perera	
4:12 PM	**A novel targeted reversed-phase LC-QqQ-MS/MS methodology reveals that ER translocon gating governs storage and membrane lipid partitioning in Saccharomyces cerevisiae** Lee Thomasson	
4:24 PM	**Investigating the Diabetic Lipidome: A DBS-Based High-Resolution Mass Spectrometry Approach** Jayden Lee Roberts	
4:36 PM	**Towards Quantitative FTIR-Based Plasma Lipid Profiling: A Design of Experiments Approach to Optimising Lipid Extraction** Satnam Kaur	
4:48 PM	**Lipidomic investigation of atherosclerotic plaque in S. pneumoniae inoculated ApoE-/- mice** Kanishka Senthilvel	
7:00 PM - 10:00 PM	**Conference Dinner** SCIEX	Dinner Reel Kitchen

Wednesday 21 October 2026

8:00 AM - 8:45 AM	**Registration**	Registration The Mezzanine
8:45 AM - 9:30 AM	**Keynote - A/Prof. Shane Ellis**	Keynote Riverside Ballroom
8:45 AM	**Next-Generation Approaches for Enhancing Spatial Lipidomics** A/Prof. Shane Ellis	
9:30 AM - 10:30 AM	**Lipid Imaging & Single-Cell Lipidomics 1** *Imaging, Spatial and Structural Lipidomics*	Session Riverside Ballroom
9:30 AM	**Lipidomic profiling of cancer cell membranes reveals signatures for therapeutic targeting** Sonia Henriques	
9:45 AM	**Assigning sn-position in glycerophospholipids by hyphenation of reversed-phase chromatography and collision-/ozone-induced dissociation mass spectrometry** Qiyu Liu	
10:00 AM	**Spatial Lipidomics Reveals Lipid Signatures Associated with Plaque Instability and Rupture in Medial-Derived Carotid Atherosclerotic Plaques** Hanane Belhoul-Fakir	
10:15 AM	**A tale of two glycerols: Uncovering the structural diversity of phosphatidylglycerol congeners through multi-dimensional analysis** Therese M. Fulloon	
10:30 AM - 11:15 AM	**Morning Tea**	Break Silver Room
11:15 AM - 12:25 PM	**Lipid Particles and Delivery Systems 1** *Lipid synthesis, delivery and metabolism*	Session Riverside Ballroom
11:15 AM	**RNA Capture and Retention in Lyotropic Lipid Mesophases** Dev Chauhan	
11:27 AM	**Targeted lipid nanoparticle delivery of plasmalogen precursors as a therapeutic strategy to restore plasmalogen levels and enhance cellular resistance to stress-mediated injury** Pukjera Jena Rungreang	
11:39 AM	**Developing Potent and Selective Ceramide Synthase 6 Inhibitors for the Treatment of Metabolic Dysfunction in the Liver** Patricia Schwarzkopf	
11:51 AM	**Inhibition of lysosomal beta glucocerebrosidase promotes de novo synthesis in a neuronal model** Jennifer Saville	
12:03 PM	**Acetaminophen-Mediated Alteration of Cardiolipin distribution in Rat Organ Tissue** HAY-YAN WANG	
12:15 PM	**COmbiNtorial Chemistry for QUantitative Standards (CONCQUEST): Quantitative Phospholipid Standards via Randomized De Novo Synthesis** Yanning Li	
12:30 PM - 2:00 PM	**Closing Lunch** Agilent Technologies	Lunch Sen5es Restaurant and Wine Bar
2:00 PM - 3:00 PM	**Closing Plenary - Prof. Aleksandra Filipovska**	Plenary Riverside Ballroom
2:00 PM	**Mitochondrial Architecture in Health and Disease: from Membrane Complexes to Organelle Networks** Prof. Aleksandra Filipovska	
3:00 PM - 3:30 PM	**Closing Address | Awards & Prizes**	Closing Riverside Ballroom